JOSHUA (JOSH) ZHONG

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EDUCATION

NEW YORK UNIVERSITY, TANDON SCHOOL OF ENGINEERING Brooklyn, NY
Master of Science in Financial Engineering Expected 05/27
GPA: No Graduate GPA - First semester of school

WEST CHESTER UNIVERSITY OF PENNSYLVANIA West Chester, PA
Bachelor of Science in Mathematics - Statistics concentration 08/24
GPA: 4.0/4.0; Dean's List (All semesters)

PROGRAMMING / TECHNICAL SKILLS

- Programming Languages: R (Tidyverse, ggplot2), Python (NumPy, pandas), C++, Git, Java, LaTeX
- Statistical & Quantitative Methods: Regression Modeling, Hypothesis Testing, Time Series Analysis, Principal Component Analysis (PCA), Clustering

COURSEWORK HIGHLIGHTS

- Mathematics & Statistics: Mathematical Statistics, Topics in Advanced Statistics, Time Series & Experimental Design, Differential Equations
- Programming: Machine Learning in Financial Engineering, Statistical Learning, C++ Programming for Financial Engineering
- Finance & Economics: Intro to Derivative Securities, Quantitative Methods in Finance

EXPERIENCE

VIEWS DIGITAL MARKETING, Norristown, PA 12/23 - 07/25
Data Analyst, AI Model Development
- Designed and configured a custom GPT model using OpenAI's ChatGPT platform to automate content generation from client website data
- Integrated structured data from Google Analytics 4 to inform model prompts and improve output relevance
- Conducted iterative testing and refinement to optimize generated content quality, reducing blog creation time by 15-20%

ACADEMIC PROJECTS

WEST CHESTER UNIVERSITY, West Chester, PA
K-Means Clustering Analysis of U.S. States [Link] 07/24
- Reduced 31 socioeconomic features to 6 principal components (~83% variance) using PCA, improving K-Means clustering quality; Within-cluster Sum of Squares (WSS) decreased by 45%, silhouette score increased from 0.23 to 0.31

Time Series Forecasting of Monthly Sunspots [Link] 01/24
- Decomposed monthly sunspot time series into trend, seasonal, and residual components via STL, revealing non-linear shifts and the ~11-year solar cycle
- Designed a rolling-origin evaluation framework with baseline benchmarking to assess forecast accuracy across varying training lengths

Modeling California Housing Markets using Multiple Linear Regression [Link] 12/23
- Engineered geospatial features by replacing raw coordinates with K-means clusters and distance-to-coast/city metrics, increasing predictive accuracy and interpretability of regression outputs
- Applied Box-Cox transformation and bootstrap resampling to correct heteroscedasticity and produce stable coefficient estimates with reliable confidence intervals
- Optimized model specification using multicollinearity and residual diagnostics, balancing predictive performance and interpretability

HONORS / AWARDS

- **Summa Cum Laude**, West Chester University, 08/24
- **Mathematical Endowment Award**, West Chester University, 08/23

EXTRACURRICULAR ACTIVITIES

- **NYU Consulting Club,** Member, 2025-Present
- **WCU American Statistical Association,** President, 2023-2024
- **WCU Student Government Association,** Senator for Student Leadership and Engagement, 2023-2024

U.S. Citizen